

SHAWAIZ KHAN

Full Stack Developer | MERN Stack & AI/RAG Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience building MERN stack and AI-powered applications, including Retrieval-Augmented Generation (RAG) systems using LangChain, Ollama, PostgreSQL + pgvector, ChromaDB, Gemini API, and Groq API. Experienced in developing secure full-stack web applications, RESTful APIs, semantic search systems, and document question-answering solutions. Currently pursuing a Bachelor of Science in Computer Science and seeking internship or entry-level software engineering opportunities.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

2024 – 2027

University of Sindh, Jamshoro · Expected Graduation: 2027

Intermediate (Pre-Medical)

2022

IBA Public School Sukkur

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6) • Python • HTML5 • CSS3

Frontend: React.js • Bootstrap • React Router

Backend: Node.js • Express.js

Databases: MongoDB • PostgreSQL • MySQL • pgvector

AI / LLM Technologies: LangChain • Ollama • ChromaDB • Gemini API • Groq API

Tools: Git • GitHub • VS Code • WordPress • WooCommerce • XAMPP

Concepts: RESTful API Development • JWT Authentication • Retrieval-Augmented Generation (RAG) • Vector Databases • Semantic Search

AI & RAG PROJECTS

RAG-Based Semantic Search System (PostgreSQL + pgvector)

Built a semantic search system that stores and indexes document embeddings directly in PostgreSQL via the pgvector extension, with local embedding and inference through Ollama models for efficient similarity search across stored content.

Technologies: Python • PostgreSQL • pgvector • Ollama

Advanced Multi-Document RAG System

Built a Retrieval-Augmented Generation system processing multiple PDF documents with incremental indexing, allowing new documents to be added without rebuilding the existing vector database.

Technologies: Python • LangChain • ChromaDB • Ollama

RAG-Based Document Question Answering System

Developed an AI-powered document question-answering application using Gemini embeddings, ChromaDB for vector storage and similarity search, and Groq LLM to generate context-aware responses from uploaded PDF documents.

Technologies: Python • Gemini API • ChromaDB • Groq API

FULL STACK WEB PROJECTS

iNotebook – Secure MERN Notes Application

Full-stack note management application with JWT-based authentication, protected CRUD operations, React Context API for state management, RESTful APIs, and a responsive Bootstrap interface.

Technologies: MongoDB • Express.js • React.js • Node.js

Shawaiz's Store – E-Commerce Website

Customized and deployed a production-ready e-commerce website using WordPress and WooCommerce, featuring product management, shopping cart, checkout, and customer account functionality.

Technologies: WordPress • WooCommerce

Live Website: shawaiz.aleemassociates.com

Student Registration System

Built a complete CRUD application with server-side validation, duplicate record prevention, and secure database operations using Express.js and MongoDB.

Technologies: Node.js • Express.js • MongoDB • REST API

Full-Stack To-Do Application

Created a task management application supporting complete CRUD functionality with MongoDB storage and asynchronous communication between frontend and backend using REST APIs.

Technologies: MongoDB • Node.js • JavaScript • REST API

NewsMonkey – React News Application

Developed a responsive news application that retrieves live headlines across multiple categories using News API, with client-side routing through React Router.

Technologies: React.js • JavaScript • Bootstrap • React Router • REST API

TextUtils – React Text Utility Application

Built a text utility application providing text transformation, word and character counting, reading-time estimation, clipboard support, live preview, and dark/light mode.

Technologies: React.js • JavaScript • Bootstrap • React Router

ACHIEVEMENTS

- Built three Retrieval-Augmented Generation (RAG) applications using LangChain, PostgreSQL + pgvector, ChromaDB, Ollama, Gemini API, and Groq API.
- Developed multiple full-stack MERN applications featuring secure authentication, RESTful APIs, and database integration.
- Built and deployed a production e-commerce website using WordPress and WooCommerce.
- Applied vector databases and modern LLM APIs to build document retrieval and semantic search solutions.

LANGUAGES

English — Professional Working Proficiency | **Urdu** — Native / Fluent